

43 Disaster PDFs(,)

Setup

Phase 1: EnvironmentPython 3.12.10, CUDA 12.4

PyTorch 2.6.0transformers 5.5.0

Detection

Phase 2: Layout AnalysisTable Transformer Detection

microsoft/table-transformerFP16, conf_t0.7

52 pages38 tables, 19 text blocks

Text
Extraction

Phase 3a: OCR ExtractionHybrid Strategy

PyMuPDFText Layer

68.4%26/38 tables

Tesseract 5.5jpn+eng

86.8%33/38 tables

Semantic
Generation

Phase 3b: Caption Generationfrom OCR Text

qwen2.5:7binstruct-q4_k_m

Success: 86.8%33/38 structured captions

Vectorization

Phase 4: Multimodal IndexEmbedding + Vector Store

hotchpotch1024-dim + Qdrant

57 blocks indexed(38 tables + 19 text)

Multimodal IndexReady for Retrieval